HW 3: Solution

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Part II:

Prove or Disprove the following claims.

1. $\{12a + 25b : a, b \in \mathbb{Z}\} = \mathbb{Z}$

   The statement above is true.
   Proof: We will show that each set is a subset of the other.

   First, let $x \in \{12a + 25b : a, b \in \mathbb{Z}\}$. Since x is the sum and product of integers, x is also an integer, so $x \in \mathbb{Z}$. Thus $\{12a + 25b : a, b \in \mathbb{Z}\} \subseteq \mathbb{Z}$.

   Now let $x \in \mathbb{Z}$. Notice that if $a = -2$ and $b = 1$ we have $12(-2) + 25(1) = 1$. Then if we multiply both sides by x, we get $(12(-2) + 25(1))x = (1)x$, or $12(-2x) + 25(x) = x$. This shows (with $a = -2x$ and $b = x$) that $x \in \{12a + 25b : a, b \in \mathbb{Z}\}$. Thus $\mathbb{Z} \subseteq \{12a + 25b : a, b \in \mathbb{Z}\}$.

   Since each set is a subset of the other, we conclude they are equal.

2. $\forall x\ \exists y, x|y$ (for 2–5 assume universe = $\mathbb{Z}$)

   The statement above is true.
   Proof: $\forall x$, let $y = x$, then $y = 1 \cdot x$, which means that $x|y$

3. $\exists y\ \forall x, x|y$

   The statement above is true.
   Proof: Let $y = 0$, then $\forall x, 0 = y = 0 \cdot x$, which means that $x|y$

4. $\exists x\ \forall y, x|y$

   The statement above is true.
   Proof: Let $x = 1$, then $\forall y, y = y \cdot 1 = y \cdot x$, which means that $x|y$

5. $\forall x,\ \forall y,\ x|y$

   The statement above is false.
   Counterexample: Let $x = 2$ and $y = 1$. Note that $2 \nmid 1$.

6. $\forall x \in \mathbb{Z}^+,\ \exists p \in P,\ p|x$ (where $\mathbb{Z}^+ = \{\text{positive integers}\}$ and $P = \{\text{primes}\}$.)

   The statement above is false.
   Counterexample: Let $x = 1$. 1 is not divisible by any prime (it is only divisible by 1 and -1, neither of which are primes). NOTE: The structure of the original statement ($\forall\ \exists$) means that in order to prove the negation ($\exists\ \forall$) you need to produce a specific x which fails for **all** p.

7. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x \neq 0 \rightarrow xy = 1$
   The statement above is true.
   Proof: $\forall x \in \mathbb{R}$, if $x \neq 0$, then we can always construct the number $\frac{1}{x} \in \mathbb{R}$. Let $y = \frac{1}{x}$; then $xy = x \cdot \frac{1}{x} = 1$

3rd{8.3}
Answer: Each bit has 2 choices and there are k bits total, so the total number of different strings that can be made is $\underbrace{2 \cdot 2 \cdot 2 \cdot \ldots \cdot 2}_{k \text{ bits}} = 2^k$

3rd{8.9}
Answer: When a rook is placed on the check board, the column and the row it stands is no longer a choice for the other rook, so there are $7 \cdot 7 = 49$ places where the other rook can be placed. And for the first rook that is being placed, all the spaces is available, which are $8 \cdot 8 = 64$ choices. So the final answer is

$$(8 \cdot 8) \cdot (7 \cdot 7) = 3136$$

3rd{8.14}
Answer: All the possibilities are listed below:
- One character: $26 + 10 = 36$
- Two characters: $36^2 = 1296$
- Three characters: $36^3 = 46,656$

...
- Eight characters: $36^8 = 2,821,109,907,456$

So that the final answer is (approximately 3 trillions)

$$36^1 + 36^2 + 36^3 + \ldots + 36^8 = 2,901,713,047,668$$

3rd{Bonus: 8.15}
Answer:

$$9^5 = 59049. \text{ Why is this correct?}$$

3rd{9.5}
Answer:

$$\frac{100!}{98!} = \frac{100 \cdot 99 \cdot 98 \cdot 97 \cdot \ldots \cdot 2 \cdot 1}{98 \cdot 97 \cdot \ldots \cdot 2 \cdot 1} = 100 \cdot 99 = 9900$$

3rd{9.6}
Answer: $100^2 < 10^{10} < 2^{100} < 100! < 100^{100}$
Explanations (optional but good practice with exponents):
$100^2 = (10^2)^2 = 10^4$ so it is less than 10^{10}.
$2^{100} = (2^{10})^{10} = 1024^{10}$ so 10^{10} is less than it.
$2^{100} = 2 \cdot 2 \cdot 2 \cdot \ldots \cdot 2$ whereas $100! = 100 \cdot 99 \cdot 98 \cdot \ldots \cdot 1$. Each have a hundred factors as written but the factors of 2^{100} are each $\leq$ the factors of 100!, except for the last one which is easily compensated for by the rest. So $2^{100} < 100!$.
Similarly, the hundred factors of 100! are each $\leq$ the hundred factors of 100^{100}.

3rd{10.3} OR 2nd{9.2}
Answer:
a. $|\{x \in \mathbb{Z} : |x| \leq 10\}| = |\{-10, -9, \ldots, -1, 0, 1, \ldots, 9, 10\}| = 21$

b. $|\{x \in \mathbb{Z} : 1 \leq x^2 \leq 2\}| = |\{-1, 1\}| = 2$

c. $|\{x \in \mathbb{Z} : x \in \emptyset\}| = |\{\}| = 0$

d. $|\{x \in \mathbb{Z} : \emptyset \in x\}| = |\{\}| = 0$
Notice that the condition here is not even syntactically correct, because x itself is not a set (even though $\emptyset$ is), so you can't say $\emptyset \in x$, which is equivalent to *set belongs to an element*. So this condition can never be satisfied.

e. $|\{x \in \mathbb{Z} : \emptyset \subseteq \{x\}\}| = |\{x \in \mathbb{Z}\}| = \infty$

f. $|2^{2^{\{1,2,3\}}}| = 2^{|2^{\{1,2,3\}}|} = 2^{2^{|\{1,2,3\}|}} = 2^{2^3} = 2^8 = 256$

g. $|\{x \in 2^{\{1,2,3,4\}} : |x| = 1\}| = |\{\{1\},\{2\},\{3\},\{4\}\}| = 4$

h. $|\{\{1,2\},\{3,4,5\}\}| = 2$

3rd{10.4} OR 2nd{9.3}
Answer:
a. $\in$ **b.** $\subseteq$ **c.** $\in$ **d.** $\subseteq$ **e.** $\subseteq$ **f.** $\subseteq$ **g.** $\in$

3rd{11.1} OR 2nd{10.1}
Answer:
a. $\forall x \in \mathbb{Z}, x$ is prime
b. $\exists x \in \mathbb{Z}, x$ is not prime and x is not composite
c. $\exists x \in \mathbb{Z}, x^2 = 2$
d. $\forall x \in \mathbb{Z}, 5|x$
e. $\exists x \in \mathbb{Z}, 7|x$
f. $\forall x \in \mathbb{Z}, x^2 \geq 0$
g. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy = 1$
h. $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x/y = 10$
i. $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy = 0$
j. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$
k. There are several options here for indicating membership in the set of people. It's fine to define such a set if you need one. Also, "time" could be quantified, or just referenced in the expression. For example if we let $P = \{$people$\}$ and $T = \{$times$\}$ then we could write
$\forall x \in P, \exists y \in P, x$ loves y at some time
$\forall x \in P, \exists y \in P, \exists t \in T, x$ loves y at time t

3rd{11.2} OR 2nd{10.2}
Answer:
a. There is an integer that is not prime.

$$\exists x \in \mathbb{Z}, x \text{ is not prime}$$

b. Every integer is a prime or a composite.

$$\forall x \in \mathbb{Z}, x \text{ is prime or } x \text{ is composite}$$

c. There is no integer whose square is 2.

$$\forall x \in \mathbb{Z}, x^2 \neq 2$$

d. There is an integer which is not divisible by 5

$$\exists x \in \mathbb{Z}, 5 \nmid x$$

e. Every integer is not divisible by 7.

$$\forall x \in \mathbb{Z}, 7 \nmid x$$

f. There is an integer whose square is negative.

$$\exists x \in \mathbb{Z}, x^2 < 0$$

g. There is an integer that, when multiplied by any integer, the result will never be 1.

$$\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy \neq 1$$

h. For every integer x, y, x/y will never be 10.

$$\forall x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x/y \neq 10$$

i. For every integer x, there is an integer y, such that xy is never 0.

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy \neq 0$$

j. There is an integer which is greater than or equal to every integer.

$$\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, y \leq x$$

k. There is a person who doesn't love anyone ever. (or, There is a person who never loves anyone.) Symbolically it is easier to clearly negate the second option above:

$$\exists x \in P, \forall y \in P, \forall t \in T, x \text{ doesn't love } y \text{ at time } t$$

3rd{11.4} OR 2nd{10.4}

Answer:

a. False **b.** True **c.** False **d.** True **e.** False **f.** True **g.** True **h.** True